

Electrodynamics: Problem Collection set 1

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Chapter 1. Electrostatics

Problem 1.1 Find the electric field of several typical distribution by integration using Coulomb's law:

1. z above one end of a straight line segment of length L that carries uniform line charge λ .
2. z above the center of a circular loop of radius r that carries a uniform line charge λ .
3. z above a square loop (side a) carrying uniform line charge λ .
4. z above the center of a flat circular disk of radius R carrying uniform surface charge σ .

Solution:

Problem 1.2 Find the electric field a distance z from the center of a spherical surface of radius R that carries a uniform charge density σ . Treat the case $z < R$ (inside) as well as $z > R$ (outside). **Then** use the result to find the field inside and outside a solid sphere of radius R carrying uniform volume charge density ρ .

Solution: